Stochastic Processes

Homework 3: First-Step Analysis and Markov Chains

Lecture 3 | 20 problems | 100 points | 2 optional bonus problems

Assigned reading

Sheldon M. Ross, *Introduction to Probability Models*, 10th ed.

- **Primary:** Chapter 4, introductory discrete-time Markov chains: states, one-step transition probabilities, transition matrices, and multistep transitions.
- **Supporting:** Lecture 3 on probability trees, first-step analysis, the Markov property, and conditional expectations of state-dependent rewards.

Instructions

Answer all 20 graded questions. Show probability-tree paths or first-step equations where requested; clearly define states and keep matrix row/column order consistent. Give exact answers when practical and round decimals to four places. A calculator or Python may check arithmetic, but show derivations. The two optional interview-style problems are *ungraded*. No state classification, recurrence, periodicity, or stationary distributions are required. No solutions are included in this handout.

I. Probability Trees and Conditional Reasoning — 12 points

Problem 1. Independent two-round game **[3 points]**

A player independently wins each round with probability 0.6 and loses with probability 0.4. Write W_i and L_i for a win and loss on round i.

(a) List the four possible two-round outcome paths.

(b) Find $P(W_1 \cap W_2)$ and $P(L_1 \cap W_2)$.

(c) Use these paths to find $P(W_2)$.

Problem 2. A dependent two-round game **[3 points]**

The first round is won with probability 0.5. After a win, the next round is won with probability 0.8; after a loss, the next round is won with probability 0.3.

(a) Draw a two-level probability tree and give the probabilities of WW, WL, LW, LL.

(b) Calculate $P(W_2)$ using the law of total probability.

(c) Find $P(W_1 \mid W_2)$.

Problem 3. Following paths for three rounds **[3 points]**

Continue the dependent game in Problem 2. Assume the next-round probabilities depend only on the immediately preceding result.

(a) Find $P(W_1 \cap L_2 \cap W_3)$.

(b) Find $P(W_1 \cap W_2 \cap W_3)$.

(c) Given that round 1 was a win, find $P(W_3 \mid W_1)$ by splitting according to round 2.

Problem 4. Conditional expected winnings **[3 points]**

Suppose the current result is W with probability 0.25 and L with probability 0.75. The next round is a win with probability 0.7 after W, or 0.4 after L. A win pays \$10 and a loss costs \$5.

(a) Find the unconditional probability that the next round is a win.

(b) Calculate the expected payoff on the next round.

(c) Explain whether independence of consecutive results was needed.

II. First-Step Analysis and Recursive Expectations 16 points

Problem 5. Waiting for a trading signal **[4 points]**

An inactive signal activates on the next day with probability 0.2 and otherwise remains inactive. The signal is currently inactive. Let T be the number of days until its *first* activation.

(a) Let $m = E[T]$. Condition on the first day to write an equation for m.

(b) Solve for $E[T]$.

(c) Calculate $P(T \leq 3)$.

Problem 6. Waiting for two consecutive heads **[4 points]**

A fair coin is flipped until HH first appears. Define state 0 to mean "no trailing head," and state 1 to mean "the last flip is H but HH has not appeared."

(a) Define m_0, m_1 as the expected additional flips until HH from states 0 and 1.

(b) Derive both first-step equations, counting the next flip.

(c) Solve for the expected number of flips starting from state 0.

Problem 7. Waiting for the pattern HT **[4 points]**

Flip a fair coin until the pattern HT appears. Use state 0 when the most recent flip is not a useful starting H, and state 1 when the most recent flip is H.

(a) Describe what happens from each state on the next H or T.

(b) Write first-step equations for the expected remaining flip counts m_0, m_1.

(c) Solve for m_0 and compare it with Problem 6.

Problem 8. Waiting for HTH **[4 points]**

A fair coin is flipped until HTH occurs. Use states 0 (no useful suffix), 1 (suffix H), 2 (suffix HT), and a completed state with zero remaining waiting time.

(a) Specify the next state after H and T from each of states 0, 1, and 2.

(b) Write the three first-step equations for m_0, m_1, m_2.

(c) Solve for the expected number of flips m_0.

III. Markov States and Transition Matrices 20 points

Problem 9. Profit/loss states [5 points]

A strategy's next day is profitable (W) with probability 0.7 following W, and with probability 0.4 following an unprofitable day (L).

(a) Define X_n and its state space.

(b) Construct the transition matrix in the order (W, L).

(c) State what p_{LW} and p_{WL} mean and check the row sums.

Problem 10. A two-state weather chain [5 points]

Tomorrow is sunny with probability 0.8 if today is sunny and with probability 0.4 if today is rainy.

(a) Write the transition matrix for state order (S, R).

(b) Draw a state-transition diagram, including both self-loops.

(c) Find the probability of the path $S \to R \to S$, conditional on starting sunny.

Problem 11. What information belongs in a state? [5 points]

Suppose the probability of tomorrow's coin outcome depends on *both* of the two most recent outcomes, rather than only today's outcome.

(a) Explain why a state recording only today's outcome need not satisfy the Markov property.

(b) Propose a four-state description that keeps the necessary information.

(c) From states HH and HT, give the two possible next states after observing the next H or T.

Problem 12. Recognizing the Markov assumption [5 points]

For a binary process, suppose

$$P(X_3 = W \mid X_2 = W, X_1 = W) = 0.8, \qquad P(X_3 = W \mid X_2 = W, X_1 = L) = 0.65.$$

Assume both conditioning events have positive probability.

(a) Can the displayed probabilities be consistent with the Markov property when the state is only $X_n \in \{W, L\}$? Explain.

(b) State the conditional-probability equality that would be required.

(c) Give one natural way to enlarge the state in order to retain the relevant history.

IV. Multistep Transitions 24 points

Problem 13. Two steps in the win/loss chain [6 points]

For states (W, L), let

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}.$$

(a) Find $P(X_2 = W \mid X_0 = W)$ by enumerating the intermediate states.

(b) Compute the entire matrix P^2.

(c) Interpret the entry in row L, column W of P^2.

Problem 14. Propagating a weather distribution **[6 points]**

A weather chain has states (S, R) and matrix

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Today is sunny with probability 0.5 and rainy with probability 0.5.

(a) Find the distribution of tomorrow's weather.

(b) Find P^2.

(c) Find the distribution two days from now and explain how it differs from the one-step distribution.

Problem 15. Three-step transitions **[6 points]**

Return to the win/loss transition matrix in Problem 13.

(a) Calculate $P^3 = P^2 P$.

(b) Find $P(X_3 = W \mid X_0 = W)$ and $P(X_3 = W \mid X_0 = L)$.

(c) Explain in words how matrix multiplication sums over intermediate paths.

Problem 16. Chapman–Kolmogorov with three states **[6 points]**

Consider the time-homogeneous Markov chain with states (A, B, C) and

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0.4 & 0.6 \end{pmatrix}.$$

(a) Verify that this is a valid transition matrix.

(b) Starting from A, calculate $P(X_2 = C \mid X_0 = A)$ by conditioning on X_1.

(c) Write the Chapman–Kolmogorov formula for $p_{AC}^{(2)}$ and identify the matching entry of P^2.

V. Conditional Expectations and Synthesis 28 points

Problem 17. Expected trading payoffs **[7 points]**

For the matrix in Problem 13, assign the next day's payoff $r(W) = \$10$ and $r(L) = -\$5$. The strategy is currently in state W.

(a) Find $E[r(X_1) \mid X_0 = W]$.

(b) Find $E[r(X_1) \mid X_0 = L]$.

(c) Use P^2 to find $E[r(X_2) \mid X_0 = W]$.

(d) Find the expected total payoff over the next two days, given $X_0 = W$, and explain whether independence is needed.

Problem 18. Weather-driven revenue and variance **[7 points]**

Use the weather matrix in Problem 14. A sunny day yields \$100 and a rainy day yields \$40. Assume today is *certainly sunny*. Let R_n be the revenue on day n.

(a) Compute $E[R_1 \mid X_0 = S]$.

(b) Compute $E[R_1^2 \mid X_0 = S]$ and $\mathrm{Var}(R_1 \mid X_0 = S)$.

(c) Using two-step probabilities, compute $E[R_2 \mid X_0 = S]$.

(d) Find $E[R_1 + R_2 \mid X_0 = S]$.

Problem 19. Reaching a target before bankruptcy **[7 points]**

A gambler starts with \$1 and on each round wins or loses \$1, each with probability 1/2. Play stops on reaching \$0 or \$3. Let h_i be the probability of reaching 3 before 0 from capital i.

(a) Give h_0 and h_3.

(b) Condition on the first move to write equations for h_1, h_2.

(c) Find the probability of reaching 3 first, starting from 1.

(d) If t_i is the expected number of additional rounds until either boundary, write and solve the equations for t_1, t_2.

Problem 20. A complete two-day trading forecast **[7 points]**

A trading regime follows the transition matrix

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} \quad \text{on states } (W, L).$$

Today is W with probability 0.5. A future W day earns \$100; a future L day loses \$50.

(a) Find $P(X_1 = W)$ using total probability.

(b) Find $P(X_2 = W)$ by conditioning on X_1.

(c) Find the expected payoff on days 1 and 2 separately.

(d) Find the expected total payoff and explain why it can be computed without assuming daily independence.

Bonus. Quant Interview Challenges optional, ungraded

Problem B1. Biased gambler's ruin (hedge-fund interview style)

A trader starts with \$2, gains \$1 with probability 0.6, and loses \$1 with probability 0.4 each round. Trading ends at \$0 or \$4. By conditioning on the first move, calculate the probability of reaching \$4 before \$0. Show the full system of equations rather than quoting a formula.

Problem B2. Three heads in a row (hedge-fund interview style)

How many flips of a fair coin are expected until HHH first appears? Define states according to whether the trailing sequence contains zero, one, or two consecutive heads. Draw or describe the first-step branches and solve the resulting expectation equations.